Formule

$\#$ permutacij $= |\{\sigma : A \xrightarrow{\sim} A\}| = n!$

$B \subseteq A \Rightarrow |A - B| = |A| - |B|$

$|A \times B| = |A||B|$

Št neurejenih izborov podmnožice k od $n = \binom{n}{k}$

multinomski: n_1 tipa 1 n_2 tipa 2, ... n_k tipa k

$\#$ permutacij neurejeno $= \dfrac{n!}{n_1! \cdot \ldots \cdot n_k!}$

① $J = \{ \underbrace{\text{juha, solata}}_{\text{predjed}}, \underbrace{\text{riba, zelenjava}}_{\text{glavna jed}}, \underbrace{\text{sladoled, torta}}_{\text{sladica}} \}$

$2 \cdot 3 \cdot 2$ je št možnih menijev

$\parallel$

12

② kup kart 52 kart. karte razdelimo na 4 igralce

vsak 1 dobi 13

1. kolko načinov lahko karte teko razdelimo
2. kolko ~~ta~~ bo imel vsak igralec karte sem enega tipa (pik, srec, kriz kara)

$$4! \cdot \binom{52}{13}\binom{39}{13}\binom{26}{13}\binom{13}{13} =$$

$$= \frac{52!\,(52-13)!\cdots}{13!\,(52-13)!\,13!\cdots 13} = \frac{52!}{(13!)^4}\,4!$$

2. $1, \vee\, 4!$

③ Konstanjevica na Kolki

imamo 702 prebivalcev

Dokaži: Najmanj dva prebivalca imata enaki začetnici (26 črk)

$$\begin{array}{r} \overset{2}{25 \cdot 25} = 625 \\ 50 \\ 125 \\ \hline 623 \end{array}$$

$702 - 625 > 0$

④ imamo 10 kart

2 rdeči, 3 zelene, 5 modrih

1). # možnosti da pri rdečih izvlečem pred prvo zeleno

_ _ _ _ _ _ _ _ _

1. 0
2. $\frac{8!}{1!\,2!\,5!}$
3. $\frac{7!}{2!\,5!} + 2 \cdot \frac{7!}{4!\,1!\,2!}$
4. R _ Z Z Z _

$$\frac{\binom{10}{4} \cdot 6}{2} = \frac{10 \cdot \overset{2}{\cancel{8}} \cdot 9 \cdot 7}{\cancel{2 \cdot 3 \cdot 4} \cdot 2} \cdot \cancel{6} = \frac{1260}{2} = 630$$

modce

$$\binom{10}{5} 4 = \frac{10!}{5!\,5!} 4 = \frac{10 \cdot 9 \cdot 8 \cdot 7}{5 \cdot 4 \cdot 3 \cdot 2} \cdot 4 =$$

$=$

ii) možnosti da bo prva rdeča kogla izvlečena pred zadnjo zeleno

$$\binom{10}{5}\cdot\binom{4}{2}\,\frac{4\cdot 3}{2}=6 \qquad \underset{\uparrow}{\text{zelena}}$$

$$\binom{10}{5}6+\binom{10}{5}3$$

$$\frac{10!}{3!5!2!}-\binom{10}{5}=\frac{10!}{5!}\left(\frac{1}{3!2!}-\frac{1}{5!}\right)=$$

$$=\frac{10!}{5!}\left(\frac{5!-\overset{3\cdot 2}{\cancel{12}}}{5!\,2!\,3!}\right)=$$

$$=\frac{10!}{5!}\,\frac{4\cdot 5-2}{5!\,2!}=\frac{10!}{5!\,5!}\,\frac{18}{2}$$

$$\parallel$$

$$\binom{10}{5}9$$

13.10

1) 3 rdeče
1 zelena
5 belih izberemo 2

i) $\Omega = ?$ $\Omega = \binom{V}{2}$ $V = \{r, r, r, z, b, b, b, b, b\}$

$\Omega = \{(r,r), (r,b), (r,z), (z,b), (z,r),$
$(b,b)\ (b,r), (b,z)\}$

ii) najprej rdeča: $\frac{3}{9} = \frac{1}{3}$

iii) $R_1 = \frac{1}{3}$ druga zelena: $\frac{8}{9} \cdot \frac{1}{8} = \frac{1}{9}$

$P(R_1 \cup Z_2) = P(R_1) + P(Z_2) - P(R_1 \cap Z_2)$

$P(R_1 \cap Z_2): \frac{3}{9} \cdot \frac{1}{8}$

$P(\dots) = \frac{4}{9} - \frac{3}{8 \cdot 9} = \frac{29}{8 \cdot 9}$

2) 2 urny: n bílých

i) $\bar{C}_{j,k} = \{$ najdeme v k-tém kroku zvlčenou krajtu $\}$

$P(\bar{C}_{j,k}) = ?$ funkce j,k,n

$$\bar{C}_{1,k} = \left(\frac{n}{n+2}\right)^{j-1} \frac{2}{n+2} \left(\frac{n+1}{n+2}\right)^{k-j-1} \left(\frac{1}{n+2}\right) =$$

$$= \frac{2 \cdot n^{j-1}}{(n+2)^{j+k-j-1+1}} (n+1)^{k-j-1} = 2n^{j-1} \frac{(n+1)^{k-j-1}}{(n+2)^k}$$

$X = k$

$$P(X,k) = P\left(\bigcup_{i=1}^{k-1} \bar{C}_{i,k}\right) = \sum_{i=1}^{k-1} P(\bar{C}_{i,k}) = \frac{2}{(n+2)^k} \sum_{j=1}^{k} n^{j-1} (n+1)^{k-j-1}$$

$$n^{j-1} (n+1)^{k-j-1} = \frac{n^j}{n(n+1)^j} (n+1)^{k-1} = \left(\frac{n}{n+1}\right)^j \frac{(n+1)^{k-1}}{n}$$

$$= \frac{2(n+1)^{k-1}}{n(n+2)^k} \frac{1 - \left(\frac{n}{n+1}\right)^k}{1 - \frac{n}{n+1}} = \frac{2((n+1)^k - n^k)}{n(n+2)^k}$$

iii) $P(\bar{C}_1 | X = k)$

$$P\left(\bigcup_{i=2}^{k} \bar{C}_{i,k}\right) = P(X=k) - P(\bar{C}_{1,k}) = \frac{2((n+1)^{k-1} - n^{k-1})}{(n+2)^k} - 2\frac{(n+1)^k}{(n+2)^k} =$$

$$= \frac{2}{n(n+2)^k} \left((n+1)^k (n+1) - n^k\right) = \frac{1}{n(n+2)^k} \left((n+1)^{k+1} - n^k\right)$$

cz: $\frac{2}{\cancel{(n+2)^k}} \left((n+1)^{k-1} \cancel{(n+2)^k} - n^{k-1}\right) =$

$$2(n+1)^{k-1} - \frac{n^{k-1}}{(n+2)^k}$$

3)

K ....konzervativec $\rightsquigarrow P(A|K) = 1$

L.... levičar $\rightsquigarrow P(A|L) = r$

$r = 0 \Rightarrow P(A) = 1$

$$P(A) = P(A|K)P(K) + P(A|L)P(L) = 1 \cdot p + (1-p)(1-r)$$

$$P(V_3|V_2) = \frac{P(V_3 \cap V_2)}{P(V_2)} = \frac{P(V_3)}{P(V_2)}$$

$$P(V_2) = P(V_2|C)P(C) + P(V_2|L)P(L) = 1 \cdot p + (1-p)(1-r)$$

$$P(V_3) = P(V_3|C)p + P(V_3|L)(1-p) = 1 \cdot p + (1-r)^2(1-p)$$

$$\Rightarrow P(V_3|V_2) = \frac{p + (1+r)^2(1-p)}{p + (1+r)(1-p)}$$

20.10

1) Mečemo kovanec, dokler ne pade najmanj eno število in en grb

$P(\{grb\}) = p \quad 0<p<1$

Naj bo $\{X=k\} = \{$prvič pade najmanj 1 število in 1 grb na koraku $k\}$

Meti so sobaj neodvisni

Raačunajte $P(X \geq k)$

$$P(X \geq k) = \sum_{k=n}^{\infty} P(X=n) = \sum_{n=k}^{\infty} \left(p^{n-1} \cdot (1-p) + (1-p)^{n-1} p \right) =$$

$$= \frac{(1-p)}{p^k} \sum_{k=0}^{\infty} p^n + \frac{p}{(1-p)^k} \sum_{n=0}^{\infty} (1-p)^n =$$

$$\frac{\cancel{(1-p)}}{p^{k-1}} \frac{1}{\cancel{1-p}} + \frac{\cancel{p}}{(1-p)^{k-1}} \frac{1}{\cancel{1-1+p}} =$$

$$= \frac{1}{p^{k-1}} + \frac{1}{(1-p)^{k-1}}$$

v prvih $n-1$ metih je samo eno

$$p^{k-1} + (1-p)^{k-1}$$

$p=0 \vee p=1 \Rightarrow p(X>k) = 0$

$$= \frac{1 \cdot (1-p)^{k-1} + p^{k-1}}{p^{k-1}(1-p^{(k-1)})} = \frac{(1-p)^{k-1} + p^{k-1}}{p^{k-1} - p^{2k-1}} \quad ???$$

2) Mečemo kovanec dokler ne dobimo ali m grbov ali n stavil

max metov: $m+n$ min metov: $\min\{m,n\}$

$P(\{grb\}) = p$

$P(\{X=k\}) =$ $\{X=k\} = p$ ker bo dobimo ali m grbov vali n stavil

m

$P(\{X=k\}) = p$ ker ali m grbov $= p^m (p-1)^{k-m}$

$+ p^{k-m}(p-1)^m$

k metov

m jih je neke vrste $k < 2m$

$$p^m (p-1)^{k-m} + p^{k-m}(p-1)^m$$

$P(X=\{\ \}) =$

$P(X=\{k-1$ tem poskusu pade $m-1$ grbov$\})$

$\{G=k\} = \{$ pade $m-1$ grbov v $k-1$ kosteh, potem pa spet pade grb $\}$

$\{S=k\} = \{$ Na $k-1$ kosteh pade $m-1$ stevil, potem pa spet stevilo $\}$

$$\binom{k-1}{m-1}\left(p^m(1-p)^{k-m} + p^{k-m}(1-p)^m\right)$$

3)

$X \sim Bin(n, \frac{1}{2})$

$$P(X=k) = \binom{n}{k}\underbrace{\left(\frac{1}{2}\right)}_{p}^{k}\left(1-\frac{1}{2}\right)^{n-k} = \binom{n}{k}\frac{1}{2^n}$$

Definiramo slučajni spremenljivki

Y za $k=0,\dots,n$ velja $P(Z) \sim P(X,Y)$

$$P(X=k,\ Y=k+1) = P(X=k)\frac{n-k}{n} \quad (a)$$

in $P(X=k,\ Y=k-1) = P(X=k)\frac{k}{n}$ (b)

in $P(X=k,\ Y=l) = 0$ če $|k-l|>1$ ali $l=k$

Razvrstimo porazdelitev slučajne spremenljivke Y

Rešitev:

Robna verjetnost

$$P(Y=l) = \sum_{k=0}^{n} P(X=k, Y=l) =$$

$$= P(X=l-1, Y=l) + P(X=l+1, Y=l)$$

$$P(Y=l) = \binom{n}{l-1}\frac{1}{2^n}\frac{n-l+1}{n} + \binom{n}{l+1}\frac{1}{2^n}\frac{l+1}{n} =$$

$$= \frac{1}{2^n}\left(\frac{n!}{(l-1)!\,(n-(l-1))!}\cdot\frac{n(l-1)}{n} + \frac{n!}{(l+1)!\,(n-(l+1))!}\cdot\frac{l+1}{n}\right)$$

$$= \frac{1}{2^n}\left(\binom{n-1}{l-1} + \binom{n-1}{l}\right) = \frac{1}{2^n}\binom{n}{l}$$

4)

Naj bodo $X \sim B(\frac{1}{2}) \Rightarrow X=0$ z verjetnostjo $\frac{1}{2}$ in $X=1$ z verjetnostjo $\frac{1}{2}$

$$Y \sim B(\tfrac{1}{2})$$

$$Z = Z(X,Y)$$

$$P(Z) = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} \quad \text{"} \in \mathbb{R}^4 \text{"}$$

$P(X=1, Y=0)$

Velja: * $0 \leq p_{ij} \leq 1$

* $\sum_{0 \leq i,j \leq 1} p_{ij} = 1$

Recimo da $P(Z) = \begin{pmatrix} \frac{1}{4}+\delta & \frac{1}{4}-\delta \\ \frac{1}{4}-\delta & \frac{1}{4}+\delta \end{pmatrix}$

kdaj sta X in Y neodvisna od δ

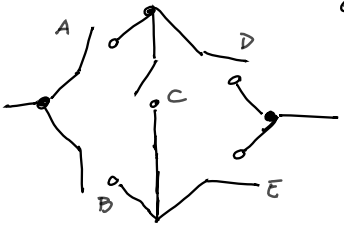
$-\frac{1}{4} \leq \delta \leq \frac{1}{4}$

Dejstvo: $X \perp Y \Leftrightarrow \det(p_{ij}) = 0$ ← neodvisna

$$\det(p_{ij}) = \left(\left(\tfrac{1}{4}-\delta\right)^2 - \left(\tfrac{1}{4}-\delta\right)^2\right) = \delta$$

$\Rightarrow X \perp Y$ če je $\delta = 0$

5) V vezje



Vsako stikala prepušča el. tok z verjetnostjo $\frac{1}{3}$ in posamezna stikala so med seboj neodvisna

A = {degeneriran tok gre preko A}

Kolikšna je verjetnost da vezje prepušča?

$$\underset{A}{\frac{1}{3}}\left(\underset{D}{\frac{1}{3}}+\underset{C}{\frac{1}{3}}\cdot\underset{E}{\frac{1}{3}}\right)+\underset{B}{\frac{1}{3}}\left(\underset{E}{\frac{1}{3}}+\underset{C}{\frac{1}{3}}\cdot\underset{D}{\frac{1}{3}}\right) \quad ??$$

$$\underset{A}{\frac{1}{3}}\left(\underset{D}{\frac{1}{3}}+\frac{1}{3}\cdot\frac{1}{3}-\frac{1}{3}\cdot\frac{1}{3}\right)+\frac{1}{3}\cdot\frac{1}{3}-\underbrace{\frac{1}{3}\cdot\frac{1}{3}}_{\text{odbijem}}=\frac{1}{9} \quad ??$$

$$\frac{2}{3}\left(\frac{1}{3}\cdot\frac{1}{3}+\frac{1}{3}\cdot\frac{1}{3}\right)+\frac{1}{3}\left(\dots\right)$$

$\uparrow$

zapleto

ست قطعه در p ژن

$$P(x=k) = P(H_c)\cdot P(x=k|H_c) + P(x=k|H_{gg})P(H_{gg}) + P(H_{go})P(x=k|H_{go})$$

$$P(x=0|H_{gg})=1$$
$$P(x=k|H_{gg})=0$$
$$P(x=0|H_{gc})=0$$
$$P(x=k|H_{gc})=P(x=k-1)$$
$$P(x=0|H_c)$$

$k=0 \Rightarrow$

$$P(x=0) = (1-p)P(x=0) + 1\cdot p^2 + p(1-p)\cdot 0$$

$$P(x=0)(1-1+p) = p^2$$
$$P(x=0) = p$$

$k\neq 0 \Rightarrow$

$$P(x=k) = P(x=k)(1-p) + \cancel{P(x=k-1)p^2} + p(p-1)P(x=k-1)$$

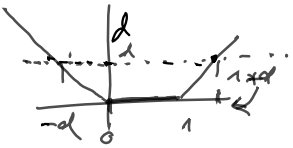
España

Naj bo $X \in U[-3,3]$ in definiramo

$$\{D = d\} = \{\max(X, [0,1]) = d\}$$

Izračunajte kumulativno porazdelitveno funkcijo za $F_d(D)$ in narišite njen graf

$$\boxed{F_D(d) = P(D \le d)} \quad D = g(X) \text{ za } g(x) = d(X, [0,1])$$



$$g(x) = \begin{cases} -x & ; x \le 0 \\ 0 & ; 0 \le x \le 1 \\ x-1 & ; 1 \le 0 \end{cases}$$

gostota:

$$f_X(x) = \begin{cases} \frac{1}{3-(-3)} = \frac{1}{6} & ; x \in [-3,3] \\ 0 & ; \text{sicer} \end{cases}$$

$$F_X(x) = \int_{-\infty}^{x} f_X(x)dx = \int_{-3}^{x} f_X(x)dx = \begin{cases} \frac{x+3}{6} & ; -3 \le x \le 3 \\ 0 & ; x < -3 \\ 1 & ; x > 3 \end{cases}$$

Kaj so dogodki $\{D \le d\}$?

$d < 0 \Rightarrow \{D \le d\} = \{\}$

$d = 0 \Rightarrow \{D \le d\} = [0,1] \quad F_D(0) = P(X \in [0,1]) = F_X(1) - F_X(0)$

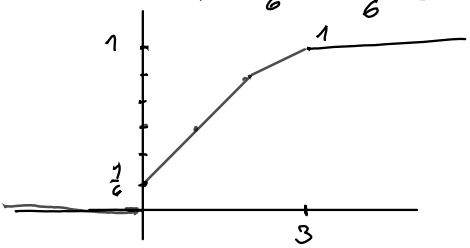
$d \in [0,2] \Rightarrow \{D \le d\} = [-d, 1+d] \quad P_D(d) = F_X(1+d) - F(-d)$

$d \in [2,3] \Rightarrow \{D \le d\} = [-d, 3] \quad P_D(d) = F_X(3) - F(-d)$

$$F_D(0) = \frac{4}{6} - \frac{3}{6} = \frac{1}{6}$$

$$F_D(d \in [0,2]) = \frac{d+4}{6} - \frac{-d+3}{6} = \frac{2d+1}{6}$$

$$F_D(d \in [2,3]) = \frac{5}{6} - \frac{-d+3}{6} = \frac{3+d}{6}$$



Vsekakor!

$$f_d = F_D'(d) = \begin{cases} 0 & : d < 0 \\ \frac{1}{3} & d \in [0,2] \\ \frac{d}{6} & d \in [2,3] \\ 0 & d > 0 \end{cases}$$

Definirajmo $X \sim$ Cauchy

$$f_X(x) = \frac{1}{\pi(1+x^2)}$$

a) Izračunajte $Y = \frac{1}{1+X^2}$

Naziv: $F_Y(y) = P(Y \leq y) = P\left(\frac{1}{1+X^2} \leq y\right) =$

$$= P\left(\frac{1}{y} \leq 1+x^2\right) = P\left(\underbrace{\sqrt{\frac{1}{y}-1}}_{X} \leq x\right) = \int_{-\infty}^{x} f_X(t)\,dt$$

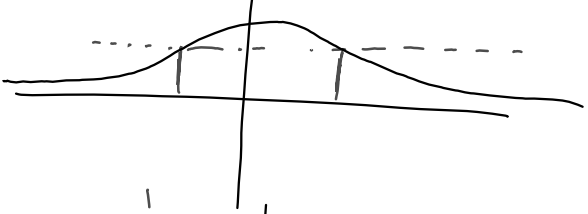
$$-x \leq$$

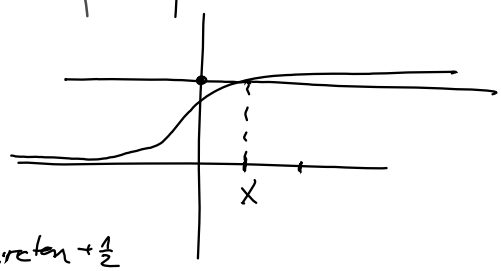
$$P\left(\sqrt{\frac{1}{y}-1} \leq x\right) = \frac{1}{\pi}\arctan(x)\Big|_{-\infty}^{x} = \frac{1}{\pi}\arctan x + \frac{1}{2}$$

$$-x \leq$$

$$P\left(\sqrt{\frac{1}{y}-1} \leq x\right) - P\left(-\sqrt{\frac{1}{y}-1} \leq x\right) =$$

$$\frac{1}{\pi}\arctan x + \frac{1}{2}$$





$$y = \frac{1}{\pi}\arctan + \frac{1}{2}$$

$$\tan\left(\pi x - \frac{\pi}{2}\right)$$

$$P(Y \le y) = P_X\left(X^2 \ge \frac{1}{y} - 1\right) = \qquad y = \frac{1}{1+X^2}$$

$$= \underset{x \ge 0}{P}\left(x \ge \sqrt{\frac{1-y}{y}}\right) + \underset{x<0}{P}\left(x \le -\sqrt{\frac{1-y}{y}}\right) =$$

$$\underset{\text{simetrija}}{= 2}\, P\left(x \ge \sqrt{\frac{1-y}{y}}\right) = 2\left(1 - F_X\left(\sqrt{\frac{1-y}{y}}\right)\right) \overset{\text{def}}{=} F_Y(y)$$

$$f_X(y) = \frac{d}{dx} F_X(x) \qquad \text{iščemo } f_Y(y)$$

$$f_X(x) = \frac{1}{\pi(1+x^2)} \qquad f_Y(y) = F_Y'(y)$$

$$\underline{\underline{f_Y(y)}} = \left(2 - 2F_X\left(\sqrt{\frac{1-y}{y}}\right)\right)' = \qquad \sqrt{\frac{1}{y}-1}$$

$$= -2 f_X\left(\sqrt{\frac{1-y}{y}}\right) \frac{-\frac{1}{y^2}}{2\sqrt{\frac{1}{y}-1}} =$$

$$= -2 \frac{1}{\pi\left(1+\frac{1-y}{y}\right)} \cdot \frac{-1}{2y^2\sqrt{\frac{1-y}{y}}} = \frac{1}{\pi} \frac{\sqrt{y}}{y^2\sqrt{1-y}}$$

$$Y = \frac{1}{1+X^2}$$

$$F_Y(y) = P_Y(Y \leq y) = P_X\left(\frac{1}{1+X^2} \leq y\right) =$$

$$\overset{?}{=} P_X\left(X^2 \geq \frac{1}{y} - 1\right) =$$ Deluje, ker je $y > 0$

ker $Y : \Omega \to (0, 1]$

in ker je $1 + x^2 > 0$

$$= P\left(X \geq \sqrt{\frac{1-y}{y}}\right) + P\left(X \leq -\sqrt{\frac{1-y}{y}}\right) \underset{\text{simetrija}}{=} 2P\left(X \geq \sqrt{\frac{1-y}{y}}\right) =$$

$$= 2\left(1 - F_X\left(\sqrt{\frac{1-y}{y}}\right)\right)$$

$$f_X(x) = \frac{1}{\pi(1+x^2)}$$

$F_X'(y) = f_X(y)$, če obstaja

$$\Rightarrow f_Y(y) = F_Y'(y) = -2F_X'\left(\sqrt{\frac{1-y}{y}}\right)$$

$$= -2 f_X\left(\sqrt{\frac{1-y}{y}}\right) \frac{1}{2}\sqrt{\frac{y}{1-y}} \cdot \frac{-y-1+y}{y^2} =$$

$$= f_X\left(\sqrt{\frac{1-y}{y}}\right)\sqrt{\frac{y}{1-y}}\,\frac{1}{y^2} =$$

$$= \frac{1}{\pi}\,\frac{1}{1+\frac{1-y}{y}}\sqrt{\frac{y}{1-y}}\,\frac{1}{y^2} =$$

$$= \frac{1}{\pi}\,\frac{y}{1}\,\frac{1}{y^2}\sqrt{\frac{y}{1-y}} = \frac{1}{\pi y}\sqrt{\frac{y}{1-y}} = \underline{\underline{\frac{1}{\pi\sqrt{y(1-y)}}}}$$

	Death		
	No	yes	
White	141	19	160
Black	149	17	166
	290	36	326

Definiramo $R: \{\text{white}, \text{black}\} \longrightarrow \mathbb{R}$

$R(\text{white}) = 0$

$R(\text{black}) = 1$

$D: \{\text{No death}, \text{yes death}\}$

$\underline{X} = (R, D)$

$\underline{U} = \{\text{Število } X = (i,j)\}$

Vprašanja?

1. $P(D=1 \mid R=i) \quad ; i \in \{0,1\}$
2. $D \perp R$? (Ali sta neodvisni spremenljivki?)

Lahko ocenimo $\hat{P}_{ij} = \dfrac{\#(X; X=(i,j))}{N=136}$

$$\hat{P}_{ij} = \begin{array}{c} \begin{pmatrix} \frac{141}{326} & \frac{19}{326} \\ \frac{149}{326} & \frac{17}{326} \end{pmatrix} \\ P(D=0) \quad P(D=1) \end{array}$$

$\hat{P}_{00} + \hat{P}_{01}$ = vrejetnost da je bel $= P(R=0)$

$= P(R=1)$

1) $P(D=1 \mid R=0) = \dfrac{P_{01}}{P_{00}+P_{01}} = 0{,}11$

$\dfrac{P_{11}}{P_{10}+P_{11}} = 0{,}10$

1. 1.12.

n gledalcev z enimi jaknami

X ... št gledalcev ki dobi nazaj jakno

$E(X) = ?$

$A_k = \{k\text{-ti obiskovalec dobi svojo jakno}$

$P(A_k) = \frac{1}{n}$

Definirajmo $I_k = \begin{cases} 1 ; & A_k \text{ se zgodi} \\ 0 ; & \neg A_k \end{cases}$

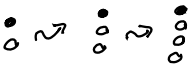
$$X = \sum_{k=1}^{n} I_k$$

$$E(X) = \sum E(I_k) = \sum \frac{1}{n} = n \cdot \frac{1}{n} = 1$$

$$\| \qquad \text{0 al: 1}$$

$$\sum + P(I_k = 1)$$

2. naloga



$$P(X=k) = \frac{k}{(k+1)!}$$

X ... st herakov igre

$$P(X \geq k) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} \cdot \ldots \cdot \frac{1}{k} \cdot \frac{1}{k+1} = \frac{1}{(k+1)!}$$

$$\cancel{\sum_{k=1}^{\infty} P(X=k)}$$

$$I_k \ldots \begin{cases} 1 & : \text{na } k \text{ se konča} \\ 0 & ; \end{cases}$$

st vseh > k

$E(X) = ?$

$$E(X) = \sum k \underbrace{\frac{k}{(k+1)!}}_{P(X=k)} = \sum \frac{k^2}{(k+1)!}$$

$$E(X) = E\left(\sum_{k=} I_k\right) = \sum E(I_k) = \sum_{k=0}^{\infty} \frac{1}{(k+1)!} = e - 1$$

3. naloga

X ima slučajna spremenljivka

$$f_X = \begin{cases} \frac{1}{\sqrt{2\pi}\, x^{\frac{3}{2}}} e^{-\frac{1}{2x}} & \text{za } x>0 \\ 0 & \text{sicer} \end{cases}$$

$Z \sim N(0,1)$ pokažite

$$F_X(x) = 2P_Z\left(Z > \frac{1}{\sqrt{x}}\right) = 2\left(1 - F_Z\left(Z \le \frac{1}{\sqrt{x}}\right)\right)$$

$$= 2\left(1 - \Phi\left(\frac{1}{\sqrt{x}}\right)\right)$$

$$F_X(x) = \int_0^x \frac{1}{\sqrt{2\pi}\, x^{\frac{3}{2}}} e^{-\frac{1}{2x}} dx = 2\int_{\frac{1}{\sqrt{x}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} =$$

$$\Phi(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} e^{-\frac{t^2}{2}}$$

$$2\int_{\frac{1}{\sqrt{x}}}^{\infty} \dots = 2\left(1 - \Phi\left(\frac{1}{\sqrt{x}}\right)\right)$$

$$u^2 = \frac{1}{x}$$

$$2u\,du = -\frac{1}{x^2} dx = dx - \frac{2u}{u}$$

$$dx = -2\frac{1}{u^3} du$$

$X \sim U(a,b) \qquad 0<a<b<1$

$y = -\frac{1}{\lambda}\ln X \qquad \lambda > 0$

$$f_X(x) = \begin{cases} \frac{1}{a-b} & x \in (a,b) \\ 0 & ; \end{cases}$$

$$P(Y \le y) = P\left(-\frac{1}{\lambda}\ln(X) \le y\right) = P(-\lambda x \le -\ln X) =$$

$$P(e^{-\lambda x} \overset{?}{\le} X) \overset{1-}{=} F_X(e^{-\lambda x}) \overset{1-}{=} \int_{-\infty}^{e^{-\lambda x}} \begin{cases} \frac{1}{a-b} & ; \\ 0 & ; \end{cases} =$$

injektivní

$-\infty = a$

$$= \begin{cases} 0 & ; e^{-\lambda x} < a \text{ nikdy} \\ \frac{e^{-\lambda x} - a}{b-a} & . e^{-\lambda x} \in (a,b) \\ 1 & ; e^{-\lambda x} \ge b \end{cases}$$

$$\begin{cases} 1 - \frac{e^{-\lambda x} - a}{b-a} & e^{-\lambda x} \in (a,b) \\ 0 \end{cases}$$

$e^{-\lambda x} \in (a,b)$

$-\lambda x \in (\ln a, \ln b)$

$\lambda x \in (-\ln b, \ln a)$

$x \in \left(-\frac{\ln b}{\lambda}, -\frac{\ln a}{\lambda}\right)$

$$P(X=k, Y=l) = \begin{cases} \underbrace{\frac{1}{2\log 2}}_{c} \frac{1}{l 2^k} & k \in \mathbb{N} \quad l \in \{1 \dots k\} \\ \text{sicer} & \end{cases}$$

$P(X=k) = ?$

$P(Y=l) = ?$

$$P(X=k) = \sum_l P(X=k, Y=l) = \sum_{l=1}^{k} \frac{1}{2\log 2} \frac{1}{l 2^k} =$$

$$\frac{1}{2^{k+1} \ln 2} \sum_{l=1}^{k} \frac{1}{l}$$

$$P(Y=l) = \sum_k P(X=k, Y=l) = \frac{1}{2\ln 2} \cdot \sum_{k=l}^{\infty} \frac{1}{2^k} =$$

$$\frac{1}{2\ln 2} \cdot \frac{1}{2^l} \frac{1 \cdot 2}{(1-\frac{1}{2}) \cdot 2} = \frac{1}{\ln 2 \, 2^l}$$

$$P(X=k)P(Y=l) = \frac{1}{2 \cdot (\ln 2)^2 2^l 2^k} \cdot \sum_{i=1}^{k} \frac{1}{i} \neq \log 2$$

ni neodvisne

Vaja

- b ptičkov
- $r < b$ označenih

Na izvedbo leta izberemo n ptičkov

X = ~~st~~ število zbranih označenih z vračanjem

Y = brez vračanja

① $var(X) = ?$

② $var(X) = \frac{b-n}{b-1} var(X)$

Nasvet: Uporabi I_k = dogodki da je k-ta ptič označen

③ kaj to pomeni za metodologijo zbiranja podatkov

Rešitev: $X \sim Bin(n, \frac{r}{b})$

$Y \sim HipGeo(n, r, b)$

Iz predavanj: $E(X) = n\frac{r}{b}$ $\qquad E(Y) = n\frac{r}{b}$

$$E(X^2) = \frac{n^2 r^2}{b^2} + n\frac{(b-r)r}{b^2}$$

$$var(X) = E(X^2) - E(X)^2 = n\frac{(b-r)r}{b^2}$$

$$E(Y^2) = \sum k P(Y^2 = k) \rightsquigarrow \sum k^2 P(Y = k)$$

$$E(Y) = \sum k P(Y = k) = \sum P(I_k = 1) =$$

~~I_k ... k-ti ptič je izbran~~

$$P(I_k \text{ je r.n. ged}) = \frac{r}{b}$$

$$E(Y) = \sum_{i=1}^{n} P(I_k) = \frac{nr}{b}$$

$$E(Y^2) = \sum k^2 P(X = k) = \sum k (kP(X = k))$$

vemo: $var(Y) = \sum var I_k + \sum_{1 \le i,j \le n,\ i \ne j} cov(I_i : I_j)$

$$cov(X, Y) = E(XY) - E(X)E(Y)$$

$$Var(Y) = \sum($$

Rešitev prejšnje naloge:

$$y=\sum_{k=1}^{n} I_k \qquad var(y)=\sum_{k=1}^{n} var I_k + \sum_{\substack{1\le i,j\le n\\ i\ne j}} cov(I_i, I_j) =$$

$$= \sum var I_k + 2\sum_{1\le i<j\le n} cov(I_i, I_j) =$$

ker $I_k \sim B(\frac{r}{b})$

$$\hookrightarrow = n\, var\, I_1 + n(n-1)\, cov(I_1, I_j)$$

Od predavanj vemo $\overbrace{E(I_k)=E(I_k^2)}^{\text{vedno velja}}=\frac{r}{b}$

$$var(I_k)=\frac{r}{b}-\left(\frac{r}{b}\right)^2=\frac{r}{b}\left(1-\frac{r}{b}\right)$$

$$cov(I_i, I_j)=E(I_i I_j)-E(I_1)E(I_j)=$$

$$= P(I_i=1, I_j=1)-\left(\frac{r}{b}\right)^2=$$

$$P(I_1=1, I_j=1)=P(I_i=1 \mid I_j=1)P(I_j)=\frac{r-1}{b-1}\,\frac{r}{b}$$

$$cov(I_i, I_j)=\frac{r}{b}\left(\frac{r-1}{b-1}-\frac{r}{b}\right)$$

$$var(y)=\frac{nr}{b}\left(1-\frac{r}{b}\right)+n(n-1)\frac{r}{b}\left(\frac{r-1}{b-1}-\frac{r}{b}\right)$$

Naj bodo $M = IA + (1-I)B$

A, B ... spremenljivke slučajne

$I \sim Ber(p)$ indikator

Predpostavimo da je I neodvisno od A in od B

Dokaži: $E(M) = pE(A) + (1-p)E(B)$

Rešitev:

$$E(M) = E(IA + (1-I)B) = E(IA) + E((1-I)B) =$$

$$= E(I)E(A) + E(1-I)E(B) = P(I=1)E(A) + P(I=0)E(B) =$$

$$= pE(A) + (1-p)E(B)$$

$\blacksquare$

n dni:

$t = 1, \dots, n$

$X_t \dots$ st klicev/minuto

$X_t = I_t Y_1 + (1 - I_t) Z_t$

- $T = U[1, \dots, n]$ je dan spremenljivka iz prve v drugo fazo

$$I_t = \begin{cases} 1 & t < T \\ 0 & t \geq T \end{cases} \qquad Y_t \sim Pois(\lambda_1) \quad Z_t \sim Pois(\lambda_2)$$

I_t je neodvisna od Y_t in od Z_t $\forall t$

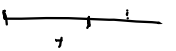
① Kaj je P? $I_t = Ber(p)$

② Najdite izraz za $E(X_t)$

③ Kaj je $E(X_t^2)$ $\qquad E(Y_1^2) = \lambda_1^2 + \lambda_1$

Rešitev:

① $p_t = P(I_t = 1) = P(t < T) = P(T > t) = \frac{n-t}{n}$



② $E(X_t) = E(I_t Y_1 + (1 - I_t) Z_t) =$

$E(I_t) E(Y_1) + E(1 - I_t) E(Z_t)$

$\frac{n-t}{n} \lambda_1 + \frac{t}{n} \lambda_2$ $\qquad var(Y_t) = \lambda_1 = var(Y_t)$

③ $E(X_t^2) = E(I_t^2 Y_1^2 + 2 I_t (1 - I_t) Y_1 Z_t + (1 - I_t)^2 Z_t^2) =$

$= \frac{n-t}{n} E(Y_1^2) + \frac{t}{n} E(Z_t^2) =$

$= (\lambda_1^2 + \lambda_1) + \frac{1}{n} (\lambda_2^2 + \lambda_2 - \lambda_1^2 - \lambda_1) =$

$= \lambda_1^2 + \lambda_1 + \frac{1}{n} (\lambda_2 - \lambda_1)(\lambda_2 + \lambda_1 + 1)$

nisan
poisalc

12. 1.